

A Comparative Study of Image Segmentation Algorithms

1. Aims

Image segmentation is the fundamental process of partitioning an image into multiple coherent regions based on pixel- or object-level characteristics [1]. Segmentation algorithms typically achieve this by detecting boundaries between distinct regions (e.g., edge-based and thresholding), by grouping features with similar properties (e.g., clustering and region growing) or a combination of both. It must be noted that in this report, segmentation is specifically referred to as multi-class segmentation rather than binary segmentation. This report aims to:

- ❖ Implement, investigate, and analyse various 2D segmentation algorithms to accurately segment the provided MRI image data. The segmentation algorithms of focus in this report include a thresholding approach, Multi-Otsu Thresholding, and two clustering techniques, namely K-Means and Mean-Shift Clustering.
- ❖ Evaluate the performance of the segmentation algorithms using an appropriate evaluation metric to facilitate the identification of the most effective segmentation algorithm relative to the ground truth images.
- ❖ Explore the disparity in segmentation performance between iteratively segmented images (2D segmentation) and simultaneous 3D segmentation.

2. Methodology

2.1 Pre-processing

Before the use of segmentation algorithms, it was crucial to conduct pre-processing steps to prepare for future segmentation. This was particularly necessary due to the presence of pixel intensity variations in regions such as the skin/scalp, which posed a challenge for accurately classifying this region into a single segment.

To tackle this challenge, active contours were used to find a boundary capable of effectively partitioning the image into two distinct regions: the brain and the combined skin/scalp and background regions. Initially, the contours' starting conditions were defined with coordinates positioned approximately between the brain and skull to locate the desired boundary. Subsequently, a binary segmentation algorithm (Binary Otsu, in this case) was utilized to separate the skin/scalp from the background using the determined threshold, resulting in the creation of a unique mask for each image. This mask could then be overlaid and combined to generate new images with the additional mask incorporated.

To elaborate, the mask's values for pixel intensity had to be manually set at a value of 50000 (the data is of float32 type). This meant the mask was slightly less dark than the background, enabling this region to be accurately identified by the segmentation algorithm.

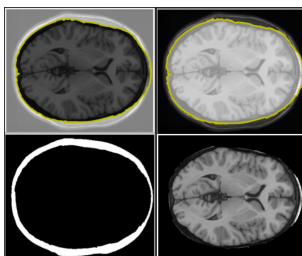


Figure 1: Binary Mask with Active Contours

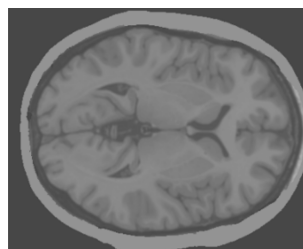


Figure 2: Image Overlayed with Mask

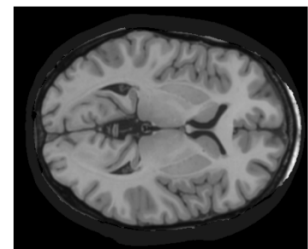


Figure 3: Final Result

2.2 Multi-Otsu Thresholding

The Multi-Otsu Thresholding algorithm is designed to automatically determine optimal thresholds for image segmentation by minimising intra-class intensity variance based on image statistics, rather than manual thresholding [2]. Multi-Otsu allows for multi-level thresholding extending upon regular Otsu Thresholding, making it particularly suitable for tasks where multiple classes need to be distinguished.

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2.3 K-Means Clustering (Outline not taught)

K-Means Clustering is another approach capable of multi-class segmentation. Initially, the algorithm sets random cluster centres, or centroids, in the feature space [3]. It then iteratively updates these centroids until convergence. In this case, the number of iterations was set to 10 to ensure convergence, as values less than 3 resulted in inconsistent segmentation performance. Pixels are then assigned to their respective centroid's class based on proximity. One notable characteristic of K-Means is its parametric nature, where users must manually specify the number of centroids.

2.4 Mean Shift Clustering (Outline not taught)

Mean Shift Clustering differs from K-Means Clustering in that it is non-parametric, so users aren't required to specify the number of clusters [3]. Instead, Mean Shift works by iteratively shifting data points towards the mode or maximum density of the data distribution. It starts with a set of data points called "kernel" or "window" and moves each point to the mean of the data points within its neighbourhood defined by the kernel. This process is repeated until convergence, where points settle into dense regions, forming clusters.

To achieve optimal results with Mean Shift Segmentation, hyperparameters within the bandwidth need to be tuned. Bandwidth controls the size of the kernel or window used to define the neighbourhood of each data point. A larger bandwidth results in a wider search radius, potentially merging neighbouring clusters, while a smaller bandwidth results in more fine-grained clusters, so the bandwidth effectively controls the number of clusters. Two hyperparameters had to be determined for the bandwidth, the number of sample and the quantile. It was found that larger samples lead to more accurately structured clusters, and the quantile was set at 0.15 to segment the image into six regions.

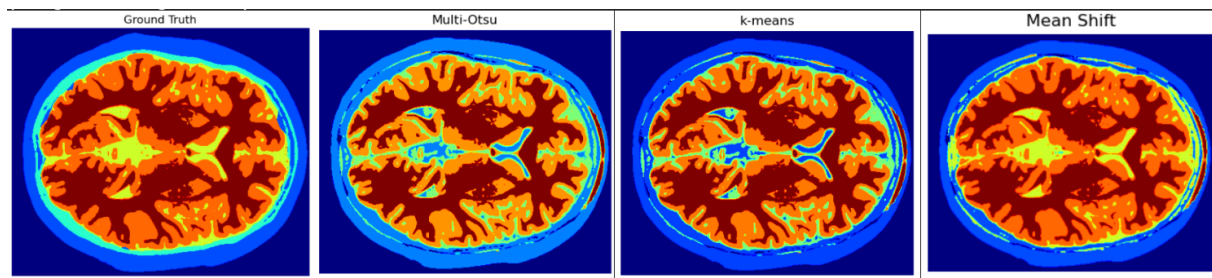


Figure 3: Examples of Segmented Images – Ground Truth, Multi-Otsu, k-means and Mean Shift.

3. Results

Table 1: F1-score results

Metric	Multi-Otsu	K-means	Mean Shift	3D Multi-Otsu
Image 1	0.9844	0.9827	0.9840	0.9807
Image 2	0.9844	0.9824	0.9840	0.9808
Image 3	0.9850	0.9832	0.9847	0.9814
Image 4	0.9846	0.9825	0.9843	0.9806
Image 5	0.9850	0.9827	0.9846	0.9805
Image 6	0.9860	0.9835	0.9855	0.9819
Image 7	0.9862	0.9843	0.9857	0.9821
Image 8	0.9869	0.9849	0.9863	0.9828
Image 9	0.9878	0.9859	0.9873	0.9833
Image 10	0.9883	0.9860	0.9879	0.9836
Average Score	0.9859	0.9838	0.9854	0.9818

Precision-recall frameworks offer a more insightful evaluation into segmentation performance compared to pixel accuracy, which is more sensitive to various factors such as outliers, class imbalance, and the number of segmented objects [4]. For example, in scenarios dominated by background classes, high accuracy may not accurately reflect the true segmentation of regions of interest (ROIs). Metrics like F1-score or Dice Coefficient within precision-recall frameworks calculate the harmonic mean of both precision and recall. Precision emphasises low false positive rates, penalising false positives common in highly class-imbalanced datasets, while recall prioritizes low false negative rates.

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In segmentation tasks aiming to accurately delineate ROIs from the background, the micro-average F1-score offers a more comprehensive evaluation of segmentation performance across all classes compared to macro-averaging. It ensures that each pixel's contribution to the final score is appropriately balanced, considering the overall segmentation accuracy. Thus, micro-average F1-score provides a holistic assessment of segmentation quality while mitigating the impact of class imbalance.

Based on the performance rankings presented in Table 1, Multi-Otsu Thresholding emerged as the top-performing segmentation algorithm, followed by Mean Shift Clustering, and lastly, K-Means Clustering. Notably, the superior performance of Mean Shift Clustering over K-Means suggests its robustness as a clustering approach. This superiority may stem from its ability to form clusters with irregular shapes and to group points within the same attraction basin.

Regarding Multi-Otsu Thresholding performing the best, it is plausible that the algorithm is particularly well-suited for the specific characteristics of the MRI images utilised in this study. The clear intensity gradients of the boundaries in these images likely facilitated the algorithm's performance. However, it's worth noting that Multi-Otsu Thresholding might encounter challenges with images containing blur or noise, where intensity variations are less distinct.

3.1 Comparing 3D and 2D Image Segmentation

Upon examining the results in Table 1, it is evident that Multi-Otsu Thresholding showed slightly lower performance in 3D image segmentation compared to its performance in iterative 2D segmentation. Interestingly, Multi-Otsu thresholding demonstrated an inherent capability for 3D segmentation without the need for specialized 3D segmentation libraries. The implementation of the 3D segmentation algorithm enabled the segmentation of the image stack in a single iteration, which contrasts with the iterative 2D approach that segmented each unique slice individually. Notably, in the case of Multi-Otsu Thresholding, thresholds were applied globally across the entire image stack, ensuring consistency throughout the dataset, whereas the iterative 2D approach required determining unique thresholds for each image.

One potential explanation for the observed decrease in segmentation performance in 3D segmentation is that the algorithm treats the entire image volume as a cohesive entity. While this global approach may offer computational advantages, it may overlook local variations and details specific to each image. The algorithm implicitly assumes a consistent spatial structure across the image stack, which may not always hold true in every scenario, although it's an appropriate assumption for the given images.

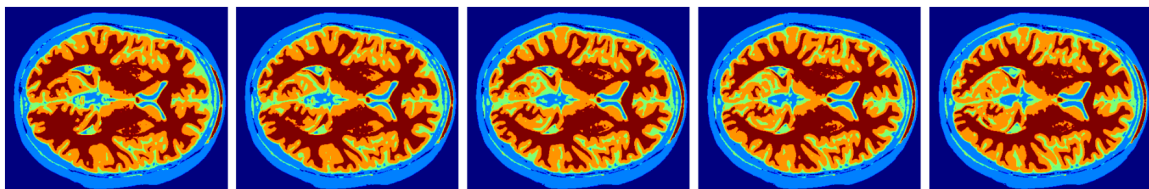


Figure 4: 2D Segmentation

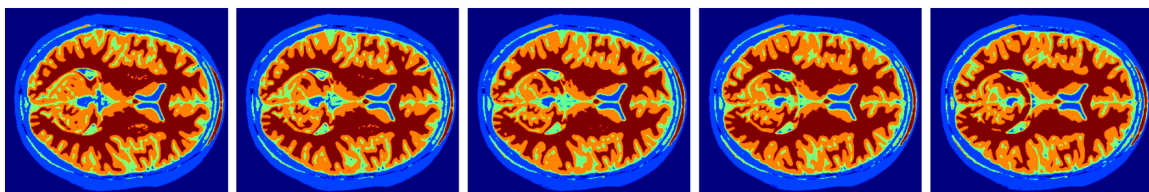


Figure 5: 3D Segmentation

Conclusion

In summary, the study compared image segmentation algorithms and found Multi-Otsu Thresholding to be the most effective, followed by Mean Shift Clustering and K-Means Clustering. Implemented pre-processing techniques, such as active contours to find boundaries for masks, proved an essential step for enhancing segmentation accuracy. The F1-score was selected as the evaluation metric over standard pixel accuracy due to its ability to handle imbalanced image regions, and it was used to assess the performance of various segmentation algorithms. Comparing 3D and iterative 2D segmentation approaches, it was observed that while 3D segmentation seems to compromise segmentation accuracy in exchange for less computational complexity.

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References

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